

AI & Data Science (AI-DS) Proper Roadmap for Engineering Students

0 → Placement Ready | Practical, Realistic & Industry-Oriented

Phase 1: Programming Foundation (1–2 Months)

- Learn Python properly: variables, loops, functions, OOP basics, file handling, exceptions.
- Practice coding regularly (LeetCode/HackerRank).
- Goal: Become comfortable solving problems in Python.

Phase 2: Math for AI-DS (2–3 Weeks)

- Focus only on useful topics: Probability, Statistics, Matrices, Basic Linear Algebra.
- Understand mean, median, variance and normalization concepts.

Phase 3: Data Science Foundation

- Learn SQL deeply: SELECT, JOIN, GROUP BY, Subqueries, Window Functions.
- Learn NumPy, Pandas, Matplotlib for data analysis and visualization.
- Do projects like Netflix analysis, IPL analysis, sales dashboards.

Phase 4: Power BI & Analytics

- Learn dashboards, KPIs, pivot tables, business reporting.
- Build resume-worthy dashboards.

Phase 5: Machine Learning (2 Months)

- Learn Linear Regression, Logistic Regression, Decision Trees, Random Forest, Clustering.
- Understand overfitting, underfitting, feature engineering and model evaluation.
- Projects: house price prediction, spam detection, customer churn.

Phase 6: Deep Learning

- Learn neural networks, CNNs, RNN basics.
- Use TensorFlow or PyTorch.
- Projects: image classifier, face recognition.

Phase 7: GenAI & LLMs

- Learn prompt engineering, embeddings, RAG, APIs.
- Build PDF chatbot, AI study assistant, resume analyzer.

Parallel Skills

- Git & GitHub
- Basic DSA
- Communication skills
- Project deployment using Streamlit/FastAPI

Correct Learning Order

Python → SQL → Pandas/EDA → Statistics → Power BI → ML → Projects → DL → GenAI → Deployment

Avoid: Random courses, tutorial bingeing, skipping projects, jumping directly to AI tools.

Target: 5 strong projects + GitHub + internship-ready skills.